Maria M. Robinson

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EMPLOYMENT

- Visiting Scholar, Psychology — present
  UNIVERSITY OF CALIFORNIA, SAN DIEGO
- Assistant Professor in Behavioral Science, Psychology — 2024-2026
  UNIVERSITY OF WARWICK
- Postdoc in Cognitive Psychology — 2019-2024
  UNIVERSITY OF CALIFORNIA, SAN DIEGO

EDUCATION

- Ph.D. in Attention and Perception — 2013-2019
  Minor in Cognitive and Quantitative Psychology
  UNIVERSITY OF ILLINOIS, URBANA-CHAMPAIGN
- M. A. in Behavior Science, focus on Decision-making — 2008-2009
  Joint research with Graduate School of Business
  UNIVERSITY OF CHICAGO
- B. A. Triple Major in Psychology (Honors), English, Philosophy — 2004-2008
  UNIVERSITY OF IOWA

RESEARCH AREAS

- Mathematical and empirical psychology
- Integrative theories of cognition
- Best practices in measurement and theory assessment in psychology

AWARDS AND FELLOWSHIPS

- Early Career Award — 2026
  PSYCHONOMIC SOCIETY
- William K. Estes Early Career Award — 2025
  SOCIETY FOR MATHEMATICAL PSYCHOLOGY
- Interdisciplinary Behavioural Research Spotlight funding (small grant scheme) — 2025
  Project title: Using Bayesian Sampling Algorithms to Think About Rumination

Collaborators: Professor Adam Sanborn, Professor Nicole Tang, Lucas Castillo
UNIVERSITY OF WARWICK

- F32 Individual National Research Service Award 2021-2024
  NATIONAL INSTITUTE OF HEALTH
- SBE Postdoctoral Research Fellowship (declined) 2021-2023
  NATIONAL SCIENCE FOUNDATION
- Graduate Dissertation Completion Fellowship 2018-2019
  UNIVERSITY OF ILLINOIS, URBANA-CHAMPAIGN
- List of Teachers Ranked as Excellent by Their Students 2015
  UNIVERSITY OF ILLINOIS, URBANA-CHAMPAIGN
- Iowa Center for Research (ICRU) Research Fellowship 2007-2008
  UNIVERSITY OF IOWA

PREPRINTS

5. Yu, K., & Robinson, M. M. (under review) Rethinking parameter invariance in cognitive modeling. `https://zenodo.org/records/23735535`
4. Yu, K., Robinson, M. M., Vanpaemel, W., Tuerlinckx, F., & Zaman, J. (revise and resubmit, PNAS) Behavioural biomarker reliability can mislead mechanistic inference in precision psychiatry. `https://osf.io/preprints/psyarxiv/dg75a_v1`
3. Zou, J., Zhang, C., Deng, H., Liu, Q., Wei, C., & Robinson, M. M. (revise and resubmit, Journal of Experimental Psychology: General) Controlled real-world object generation for visual memory experimentation and theory-building. `https://osf.io/preprints/psyarxiv/jz7uf_v3`
2. Brady, T. F. & Robinson, M. M. (under review) Predicting visual working memory errors with a priori measures of representational geometry from human and artificial systems. `https://osf.io/preprints/psyarxiv/k3xu6_v1`
1. Robinson, M. M. & Brady, T. F. (in preparation) Simple readout, complex behaviour via a generative model of confidence. `https://osf.io/preprints/osf/6zx9v_v1`

JOURNAL PUBLICATIONS

32. Robinson, M. M. & Brady, T. F. (2026). Author response to "Testing the reproducibility, replicability and robustness of Robinson and Brady (2023)". *Nature Human Behaviour*.
31. Brady, T. F., Poungtubtim, C., & Robinson, M. M. (2026). The TCC-ensemble model predicts full feature distribution reports: Commentary on Khvostov et al., (2025). *Psychonomic Bulletin and Review*.
30. Yu, K., & Robinson, M. M. (2026). Prediction, risk and illusion. *Computational Brain and Behavior*.

29. Miao, A., Brady, T. F., & Robinson, M. M., (2026). Bridging Bayesian and representational theories of memory to predict memory bias. *Journal of Experimental Psychology: General.*

28. Robinson, M. M., Williams, J. R., Wixted, J. E., & Brady, T. F. (2025). Zooming in on what counts as core and auxiliary in theory assessment: A case study on recognition models of visual working-memory. *Psychonomic Bulletin and Review.*

    Featured *Psychonomic Society* content: `https://featuredcontent.psychonomic.org/a-case-study-on-the-challenges-of-theory-testing-in-psychology/`

27. Brady, T. F., Robinson, M. M., & Williams, J. R. (2024). Noisy and hierarchical visual memory across time scales. *Nature Reviews Psychology.*

26. Robinson, M. M., DeStefano, I., Vul, E., & Brady, T. F. (2023). Local but not global graph theoretic measures of semantic networks generalize across tasks. *Behavior Research Methods.*

25. Robinson, M. M., DeStefano, I., Vul, E. & Brady, T. F. (2023). How do people build up visual memory representations from sensory evidence? Revisiting two classic models of choice. *Journal of Mathematical Psychology.*

24. Robinson, M. M., & Brady, T. F. (2023). A quantitative model of ensemble perception as summed activation in feature space. *Nature Human Behavior.*

23. Brady, T. F., Robinson, M. M., Williams, J. R., & Wixted, J. (2023). Measuring memory is harder than you think: How to avoid problematic measurement practices in memory research. *Psychonomic Bulletin and Review.*

22. Robinson, M. M., & Steyvers, M. (2022). Linking computational models of two core tasks of cognitive control. *Psychological Review.*

21. Williams, J.*, Robinson, M. M.*, & Brady, T. F. (2022). There is no theory-free measure of "swaps" in visual working memory experiments. *Computational Brain and Behavior.*

20. Williams, J., Robinson, M. M., Schurgin, M., Wixted, J. T., & Brady, T. F. (2022). You can't count how many items people remember in working memory: The importance of signal detection-based measures for understanding change detection performance. *Journal of Experimental Psychology: Human Perception and Performance.*

19. Goldenberg, A., Schone, J., Huang, Z., Sweeny, T. D., Levari, D., Ong, D. C., Zaki, J., Brady, T. F., Robinson, M. M., & Gross, J. J. (2022). Amplification in the evaluation of emotional expressions over time. *Nature Human Behavior.*

18. Regenwetter, M., & Robinson, M. M. (2022). Reply to commentaries: Why should we worry about scientific conjunction fallacies? *Decision.*

17. Regenwetter, M., Robinson, M. M., & Wang, C. (2022). Four internal inconsistencies in Tverksy and Kahneman's (1992) Cumulative Prospect Theory paper: A case study in ambiguous theoretical scope. *Advances in Methods and Practices in Psychological Science.*

16. Regenwetter, M., Robinson, M. M., & Wang, C. (2022). Are you an exception to your favorite decision theory? Behavioral decision research is a huge Linda problem! *Decision.*

15. Lively, Z., Robinson, M. M., & Benjamin, A. S. (2021). Memory strength reveals qualitative changes in interactions between items in visual working memory. *Psychological Science.*

14. Akan, M., Robinson, M. M., Mickes, L., Wixted, J. T., & Benjamin, A. S. (2020). The effect of lineup size on eyewitness identification accuracy. *Journal of Experimental Psychology: Applied.*

13. Robinson, M. M., Benjamin, A. S., & Irwin, D. E. (2020). Is there a K in capacity? Evaluating the discrete-slot model of visual short-term memory. *Cognitive Psychology.*

12. Regenwetter, M., & Robinson, M. M. (2019). Tutorial: Nuisance or substance? Leveraging heterogeneity of preferences. *The Spanish Journal of Psychology.*

11. Robinson, M. M., & Irwin, D. E. (2019). Are there two visual short-term memory stores? A state-trace analysis. *Journal of Mathematical Psychology.*

10. Regenwetter, M. & Robinson, M. M. (2018). The construct-behavior gap revisited: Reply to Hertwig and Pleskac. *Psychological Review.*

9. Robinson, M. M., Clevenger, J., & Irwin, D. E. (2018). The action is in the task-set not in the action. *Cognitive Psychology.*

8. Irwin, D. E. & Robinson, M. M. (2018). How post-saccadic target blanking affects the detection of stimulus displacements across saccades. *Vision Research.*

7. Regenwetter, M., & Robinson, M. M. (2017). The construct-behavior gap in behavioral decision research: A challenge beyond replicability. *Psychological Review.*

6. Robinson, M. M., & Irwin, D. E. (2017). Conscious error perception: The impact of response interference from a secondary task. *Attention, Perception, & Psychophysics.*

5. Irwin, D. E., & Robinson, M. M. (2016). Perceiving a continuous visual world across voluntary eye blinks. *Journal of Experimental Psychology: Human Perception and Performance.*

4. Robinson, M. M., & Irwin, D. E. (2016). Shifts of attention bias awareness of voluntary and reflexive eye movements. *Experimental Brain Research.*

3. Irwin, D. E., & Robinson, M. M. (2015). Detection of stimulus displacements across saccades is capacity limited and biased in favor of the saccade target. *Frontiers in Systems Neuroscience.*

2. Irwin, D. E., & Robinson, M. M. (2014). Perceiving stimulus displacements across saccades. *Visual Cognition.*

1. Robinson, M. M., & Morsella, E. (2014). The subjective effort of everyday mental tasks: Attending, assessing, and choosing. *Motivation and Emotion.*

CHAPTERS

1. Robinson, M. M., Poehlman, T. A., & Morsella, E. (2012). The subjective aspects of agency, cognitive control, and self-regulation: Findings from the action and consciousness laboratory. *Consciousness: States, Mechanisms, & Disorders.*

CONFERENCE TALKS AND POSTERS

25. Grigorev, D., & Robinson, M. M. (2026, November). Characterizing the structure of domain-generality in metacognitive confidence. *Talk presentation for J. Frank Yates Student Award, Psychonomics Society Conference*

24. Robinson, M. M., & Yu, K. (2026, July). Flexibility and misspecification in invariance testing: Comparing two approaches. *Keynote, Society for Mathematical Psychology*

23. Yu, K. & Robinson, M. M. (2026, July). Rethinking parameter invariance tests in congitive modeling. *Talk Presentation at Society for Mathematical Psychology*

22. Robinson, M. M., Zou, S. & Wei, C. (2025, May). Using AI generated real-world objects to uncover the structure of visual memory. *Poster Presentation at the Proceedings of the Annual Meeting of the Vision Sciences Society Meeting.*

21. Robinson, M. M. & Brady, T. F. (2025, May). Memory strength as accumulated activation over time: Unifying temporal order and repetition effects. *Poster Presentation at the Proceedings of the Annual Meeting of the Vision Sciences Society Meeting.*

20. Miao, A., Brady, T. F. & Robinson, M. M. (2025, May). Bridging Bayesian and Representational Theories of Memory to Predict Memory Biases. *Talk Presentation at the Proceedings of the Annual Meeting of the Vision Sciences Society Meeting.*

19. Brady, T. F., & Robinson, M. M. (2024, May). Phantom memories: Enhancing memory through competition with imaginary items. *Poster presented at the Proceedings of the Annual Meeting of the Vision Sciences Society Meeting.*

18. Robinson, M. M. (2023, November). Formally linking computational models of attentional control tasks can help validate measures and constrain theory. *Talk presented at the Attentional control as a psychometric construct: Challenges and responses, Symposium, Psychonomic Society Conference.*

17. Robinson, M. M., & Brady, T. F. (2023, November). Metacognition as direct readout. *Talk presented at the Psychonomic Society Conference.*

16. Brady, T. F., & Robinson, M. M. (2023, May). Visual similarity structure a priori predicts memory errors for novel high-dimensional face stimuli. *Poster presented at the Proceedings of the Annual Meeting of the Vision Sciences Society Meeting.*

15. Robinson, M. M., DeStefano, I, Vul, E., & Brady, T. F. (2023, May). Building up visual memories from sensory evidence. *Poster presented at the Proceedings of the Annual Meeting of the Vision Sciences Society Meeting.*

14. Robinson, M. M., Wixted, J., & Brady, T. F., (2022, November). What conditions optimize memory and why? Reconciling the conflicting effects of similarity and correlated noise. *Talk presented at the Psychonomics Society Conference.*

13. Robinson, M. M., DeStefano, I. C., Brady, T. F., & Vul, E., (2022, July). Robustness of graph theoretic representations of semantic networks. *Poster presented at the Proceedings of the Annual Meeting of the Cognitive Science Society.*

12. Robinson, M. M., Williams, J. R., & Brady, T. F. (2022, May). Evaluating models of visual working-memory in change detection: Discrete-slots or non-diagnostic data? *Poster presented at the Vision Sciences Society Meeting.*

11. Robinson, M. M., Brady, T. F., Vul, E., & DeStefano, I. C. (2021, July). Revisiting the connection between the Luce's choice rule and signal detection theory. *Talk presented at the Virtual Conference of the Society of Mathematical Psychology.*

10. Robinson, M. M., & Brady, T. F. (2021, May). A quantitative model of ensemble perception as summed activation in feature space. *Talk presented at the Virtual Vision Sciences Society Meeting.*

9. Robinson, M. M., Wixted, J. T., & Brady, T. F. (2020, June). Decision strategy matters: Different testing procedures can change decision strategies and lead to spurious effects on estimates of visual working memory sensitivity. *Poster presented at the Virtual Vision Sciences Society Meeting.*

8. Robinson, M. M., Irwin, D. E., & Benjamin, A. S. (2019, November). Is there a K in capacity? Evaluating the discrete-slot model of visual short-term memory. *Poster presented at the 60th Annual Meeting of the Psychonomic Society, Montreal, Canada.*

7. Regenwetter, M., & Robinson, M. M. (2017, July). A context dependent random-preference approach to the description-experience gap. *Talk presented at the Annual Meeting of the Society for Mathematical Psychology, the European Mathematical Psychology Group of the International Conference on Cognitive Modelling, Warwick.*

6. Irwin, D. E., & Robinson, M. M. (2016, November). How post-saccadic target blanking affects the detection of stimulus displacements across saccades. *Poster presented at the 57th Annual Meeting of the Psychonomic Society, Boston, MA.*

5. Robinson, M. M., & Irwin, D. E. (2016, November). Reexamining the effect of action on attentional deployment in visual search. *Poster presented at the 24th Annual Workshop on Object Perception, Attention, & Memory, Boston, MA.*

4. Robinson, M. M., & Regenwetter, M. (2015, November). Description-experience gap or construct-data gap? *Talk presented at the Annual Meeting of the Society for Judgment and Decision Making, Chicago, IL.*

3. Robinson, M. M., & Irwin, E. D. (2015, November). Conscious error perception: The impact of response interference from a secondary task. *Poster presented at the 56th Annual Meeting of the Psychonomic Society, Chicago, IL.*

2. Krisst, L. C., Robinson, M. M., & Morsella, E. (2013, May). Revisiting Libet's studies: Effect of volition on time perception and response interference. *Poster presented at the Annual Meeting of the Society for Cognitive Neuroscience, San Francisco, CA.*

1. Robinson, M. M., & Morsella, E. (2012, March). The subjective cost of everyday thinking (comparing, attending, and choosing): Implications for social cognition research. *Poster presented at the Annual Meeting of the Society for Personality and Social Psychology, San Diego, CA.*

TEACHING POSITIONS

- Co-lecturer (80%), Models of Choice, MSc Course 2025
  UNIVERSITY OF WARWICK
- Co-lecturer (50%), Judgment and Decision-Making Final Year Course 2024, 2025
  UNIVERSITY OF WARWICK
- Co-lecturer (20%), Statistics, MSc Course 2024
  UNIVERSITY OF WARWICK
- Laboratory Instructor, Introductory Psychology 2017
  UNIVERSITY OF ILLINOIS, URBANA-CHAMPAIGN
- Laboratory Instructor, Introductory Psychology 2015
  UNIVERSITY OF ILLINOIS, URBANA-CHAMPAIGN
- Adjunct Instructor of Statistics in Health and Behavioral Science 2013
  DOMINICAN UNIVERSITY OF CALIFORNIA, SAN RAFAEL

ADMINISTRATIVE POSITIONS

- Member of University's Humanity Social Science Ethics Committee 2024-2026
  UNIVERSITY OF WARWICK
- Admission Tutor, Economics, Psychology and Philosophy BA/BSc program 2025-2026
  UNIVERSITY OF WARWICK

EDITORIAL POSITIONS

- Associate Editor 2026
  Journal of Experimental Psychology: Learning, Memory and Cognition
- Guest Editor for Special Issue on Tutorials 2025-2026
  Behavior Research Methods

RESEARCH POSITIONS

- Visiting Research Associate, Action and Consciousness Laboratory 2010-2013
  SAN FRANCISCO STATE UNIVERSITY
- Undergraduate Research Scholar in Philosophy 2007-2008
  UNIVERSITY OF IOWA
- Undergraduate Research Assistant in Psychology 2006-2008
  UNIVERSITY OF IOWA